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Industrial Internship Report on

” IoT-Based Weather Reporting System using ESP32”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was This report presents the work carried out during the Industrial Internship organized by Upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT). The internship duration was 6 weeks and focused on solving a real-world problem using IoT technologies. The project developed during this internship is an **IoT-Based Weather Reporting System** that monitors environmental parameters such as temperature and humidity. The system initially used simulation through Node-RED and later integrated real hardware using ESP32 and DHT sensor. The collected data is transmitted using the MQTT protocol and displayed on a Node-RED dashboard in real-time. The system also includes alert mechanisms and data logging features. This internship provided hands-on experience in IoT system design, embedded programming, and real-time data visualization. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface 3

2 Introduction....4

2.1 About UniConverge Technologies Pvt Ltd....4

2.2 About upskill Campus....8

2.3 Objective 10

2.4 Reference 10

2.5 Glossary....10

3 Problem Statement....**Error! Bookmark not defined.**

4 Existing and Proposed solution ....**Error! Bookmark not defined.**

5 Proposed Design/ Model....13

5.1 High Level Diagram (if applicable) 13

5.2 Low Level Diagram (if applicable)....13

5.3 Interfaces (if applicable)....13

6 Performance Test 14

6.1 Test Plan/ Test Cases ....**Error! Bookmark not defined.**

6.2 Test Procedure....**Error! Bookmark not defined.**

6.3 Performance Outcome....**Error! Bookmark not defined.**

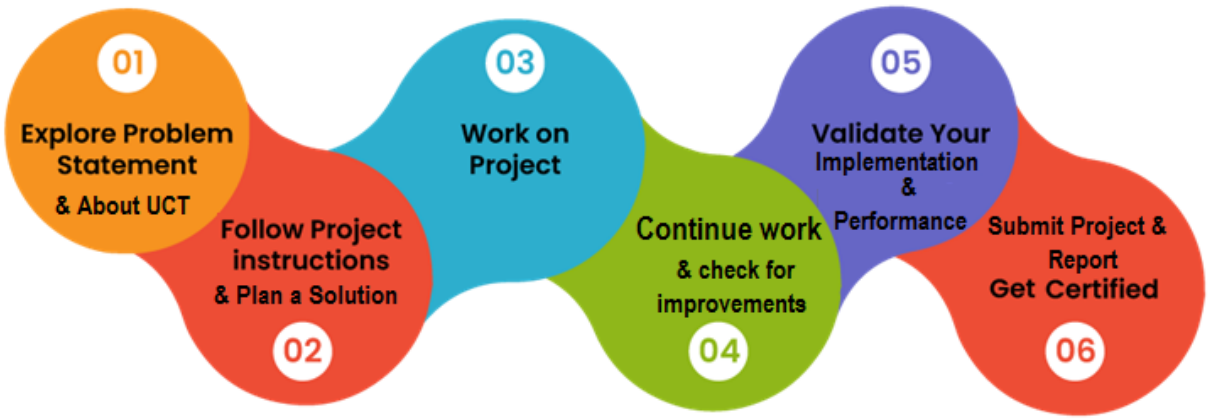
7 My learnings....15

8 Future work scope 16

1 Preface

This internship played a significant role in enhancing my technical knowledge and practical skills in the field of Embedded Systems and IoT. Over a period of six weeks, I worked on designing and implementing a real-time Weather Monitoring System.

The internship program was well-structured and provided step-by-step guidance starting from simulation to real hardware implementation. It helped me understand how industrial IoT systems are developed and deployed.



I would like to thank Upskill Campus, UniConverge Technologies, and my mentors for their continuous support and guidance throughout the internship.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

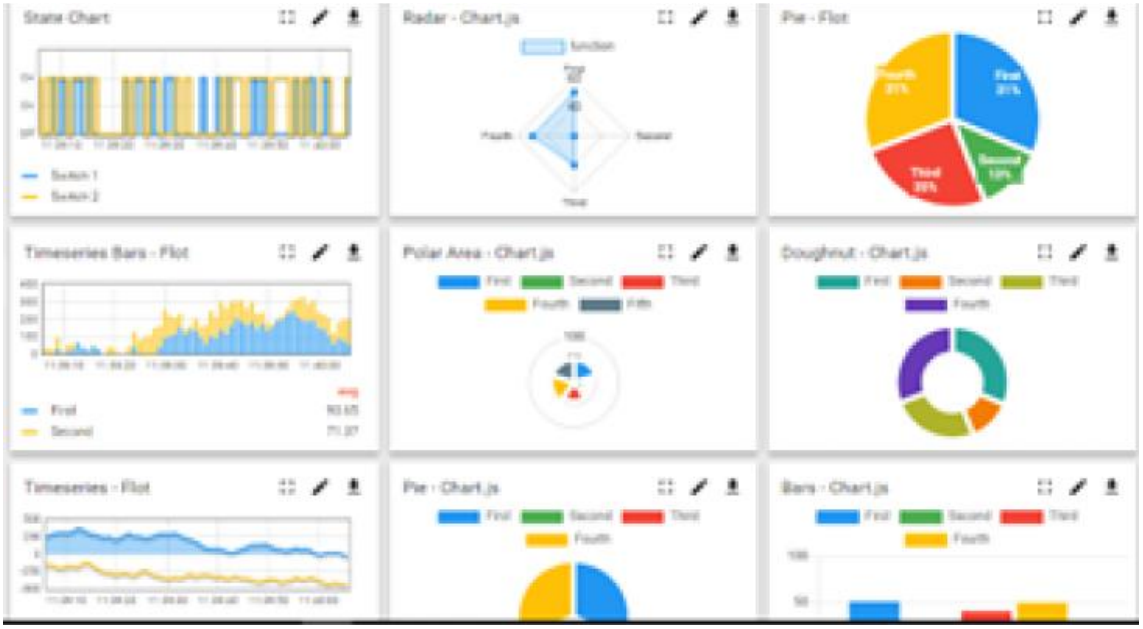
i. UCT IoT Platform (uct Insight)

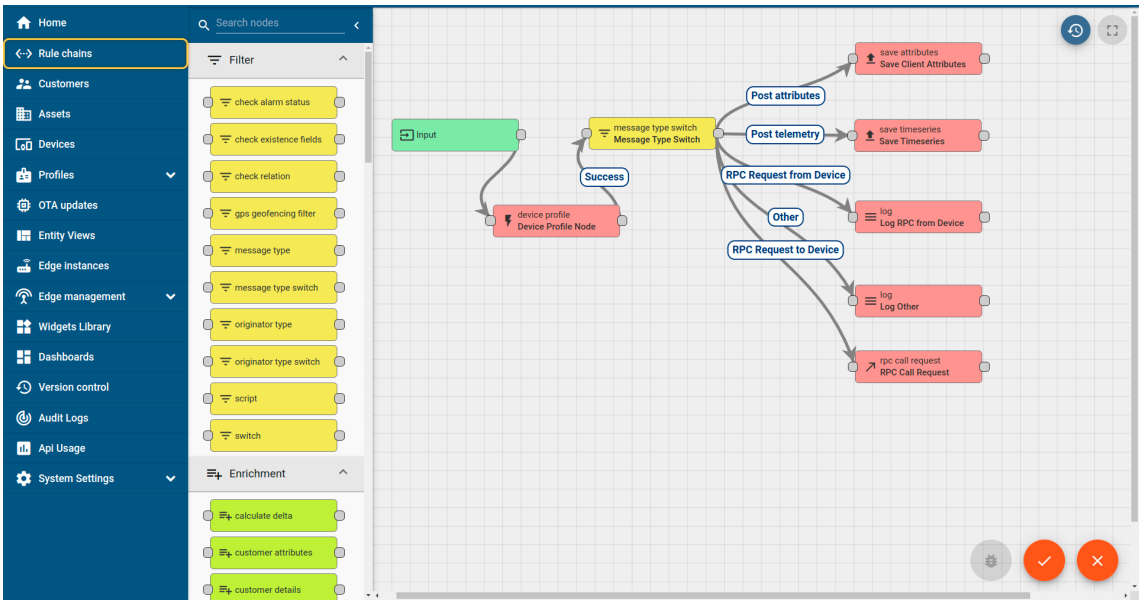
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

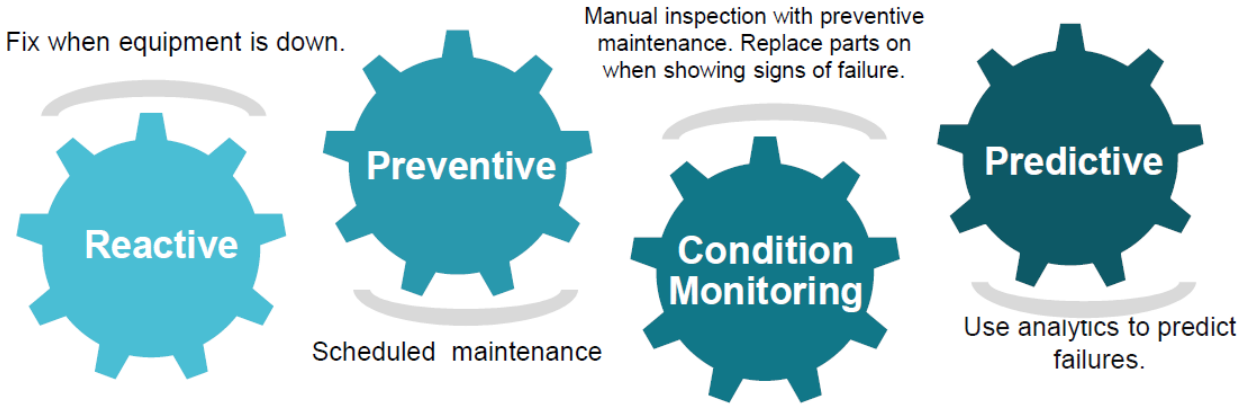
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

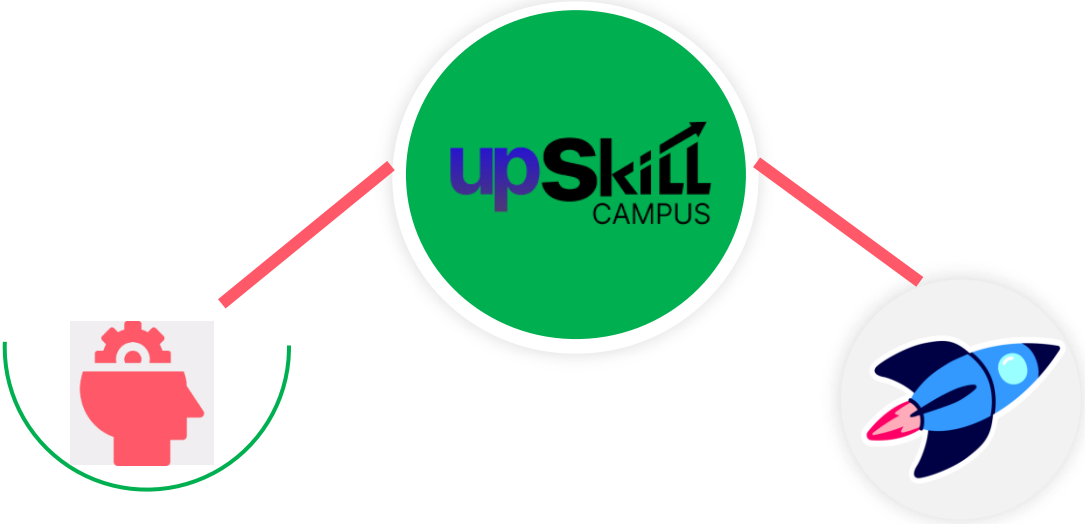
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

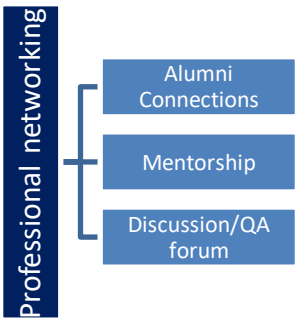
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

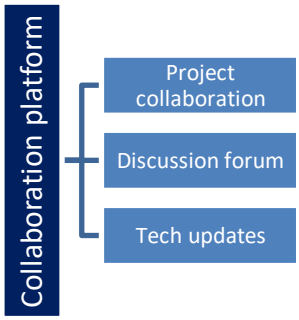


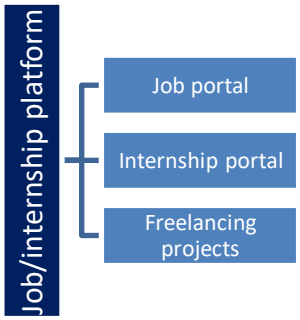
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The main objectives of this internship were:

- To gain practical knowledge of IoT systems
- To solve real-world engineering problems
- To improve technical and communication skills
- To understand industrial workflows and development processes

2.5 Reference

[1] Node-RED Official Documentation
[2] MQTT Protocol Documentation
[3] ESP32 Datasheet

2.6 Glossary

Terms	Acronym
Iot	Internet of Things
MQTT	Message Queuing Telemetry Transport
ESP32	Microcontroller with WiFi
DHT11	Temperature & Humidity Sensor

3 Problem Statement

The aim of this project is to design and implement a system that can monitor environmental conditions such as temperature and humidity in real-time.

The system should:

- Collect data from sensors
- Transmit data wirelessly
- Display real-time data on a dashboard
- Generate alerts when thresholds are exceeded

4 Existing and Proposed Solution

4.1 Existing Solutions

- Expensive weather monitoring systems
- Limited accessibility
- Lack of real-time visualization

4.2 Proposed Solution

- Low-cost IoT-based system using ESP32
- Real-time monitoring using MQTT protocol
- User-friendly dashboard using Node-RED

4.3 Value Addition

- Real-time alerts
- Data logging
- Scalable design

4.4 Code submission

https://github.com/tejastambe123/upskillcampus

5 Proposed Design/ Model

5.1 High Level Diagram

Sensor (DHT11) → ESP32 → MQTT Broker → Node-RED → Dashboard

final outcome.

5.2 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.3 Low Level Diagram (if applicable)

5.4 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

6.1 Constraints

- Network delay
- Sensor accuracy
- Power consumption

6.2 Test Plan

- Continuous data transmission every 5 seconds
- Alert triggering when temperature > 35°C

6.3 Test Procedure

- Run system continuously
- Monitor dashboard updates
- Check alert system

6.4 Performance Outcome

- Stable real-time data transmission
- Accurate sensor readings
- Reliable system performance

7 My learnings

- Learned IoT system architecture
- Understood MQTT communication
- Gained experience with ESP32 programming
- Developed dashboard using Node-RED
- Improved debugging and problem-solving skills

8 Future work scope

- Mobile app integration
- Cloud storage (AWS/Firebase)
- AI-based prediction system
- Integration of multiple sensors
- Advanced alert system (SMS/Email)